

INFT-1205 – Data Communications & Networking II

Capstone Project

Routing Documentation

Institution: Durham College

By: Group 1

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Document Overview

This document serves as brief documentation of the static route configurations on the DatacoreRouter, RogersRouter, and BellRouter.

1. show ip route – DatacoreRouter – Primary ISP

The routing tables shows the directly connected routes for all the sub interfaces. It also includes the default static route for the Primary Rogers ISP. The Backup Bell ISP route does not display unless the link to the Primary Rogers ISP is down.

```
DatacoreRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected
```

Gateway of last resort is 101.1.1.2 to network 0.0.0.0

```
S*    0.0.0.0/0 [1/0] via 101.1.1.2
      10.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
C      10.160.0.0/25 is directly connected, GigabitEthernet0/0/0.10
L      10.160.0.1/32 is directly connected, GigabitEthernet0/0/0.10
C      10.160.0.128/26 is directly connected, GigabitEthernet0/0/0.20
L      10.160.0.129/32 is directly connected, GigabitEthernet0/0/0.20
C      10.160.0.192/26 is directly connected, GigabitEthernet0/0/0.30
L      10.160.0.193/32 is directly connected, GigabitEthernet0/0/0.30
C      10.160.1.0/26 is directly connected, GigabitEthernet0/0/0.40
L      10.160.1.1/32 is directly connected, GigabitEthernet0/0/0.40
C      10.160.1.64/26 is directly connected, GigabitEthernet0/0/0.60
L      10.160.1.65/32 is directly connected, GigabitEthernet0/0/0.60
C      10.160.1.128/26 is directly connected, GigabitEthernet0/0/0.70
L      10.160.1.129/32 is directly connected, GigabitEthernet0/0/0.70
C      10.160.1.192/27 is directly connected, GigabitEthernet0/0/0.50
L      10.160.1.193/32 is directly connected, GigabitEthernet0/0/0.50
C      10.160.1.224/27 is directly connected, GigabitEthernet0/0/0.80
L      10.160.1.225/32 is directly connected, GigabitEthernet0/0/0.80
      101.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C      101.1.1.0/30 is directly connected, Serial0/1/0
L      101.1.1.1/32 is directly connected, Serial0/1/0
C      101.1.2.0/30 is directly connected, Serial0/1/1
L      101.1.2.1/32 is directly connected, Serial0/1/1
```

2. show ip route – DatacoreRouter – Backup ISP

The routing table now displays the Backup Bell ISPs default static route, with an administrative distance of 10.

```
DatacoreRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is 101.1.2.2 to network 0.0.0.0

S*    0.0.0.0/0 [10/0] via 101.1.2.2
      10.0.0.0/8 is variably subnetted, 16 subnets, 4 masks
C      10.160.0.0/25 is directly connected, GigabitEthernet0/0/0.10
L      10.160.0.1/32 is directly connected, GigabitEthernet0/0/0.10
C      10.160.0.128/26 is directly connected, GigabitEthernet0/0/0.20
L      10.160.0.129/32 is directly connected, GigabitEthernet0/0/0.20
C      10.160.0.192/26 is directly connected, GigabitEthernet0/0/0.30
L      10.160.0.193/32 is directly connected, GigabitEthernet0/0/0.30
C      10.160.1.0/26 is directly connected, GigabitEthernet0/0/0.40
L      10.160.1.1/32 is directly connected, GigabitEthernet0/0/0.40
C      10.160.1.64/26 is directly connected, GigabitEthernet0/0/0.60
L      10.160.1.65/32 is directly connected, GigabitEthernet0/0/0.60
C      10.160.1.128/26 is directly connected, GigabitEthernet0/0/0.70
L      10.160.1.129/32 is directly connected, GigabitEthernet0/0/0.70
C      10.160.1.192/27 is directly connected, GigabitEthernet0/0/0.50
L      10.160.1.193/32 is directly connected, GigabitEthernet0/0/0.50
C      10.160.1.224/27 is directly connected, GigabitEthernet0/0/0.80
L      10.160.1.225/32 is directly connected, GigabitEthernet0/0/0.80
      101.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      101.1.2.0/30 is directly connected, Serial0/1/1
L      101.1.2.1/32 is directly connected, Serial0/1/1
```

3. show ip route – RogersRouter

The RogersRouter is configured with a default static route, directed back at the Datacore Router.

```
RogersRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
        n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT D1A
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        H - NHRP, G - NHRP registered, g - NHRP registration summary
        o - ODR, P - periodic downloaded static route, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
        & - replicated local route overrides by connected

Gateway of last resort is 101.1.1.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 101.1.1.1
      101.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      101.1.1.0/30 is directly connected, Serial0/1/1
L      101.1.1.2/32 is directly connected, Serial0/1/1
```

4. show ip route – BellRouter

The BellRouter is configured with a default static route, directed back at the Datacore Router.

```
BellRouter#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override

Gateway of last resort is 101.1.2.1 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 101.1.2.1
      101.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      101.1.2.0/30 is directly connected, Serial0/0/0
L      101.1.2.2/32 is directly connected, Serial0/0/0
```